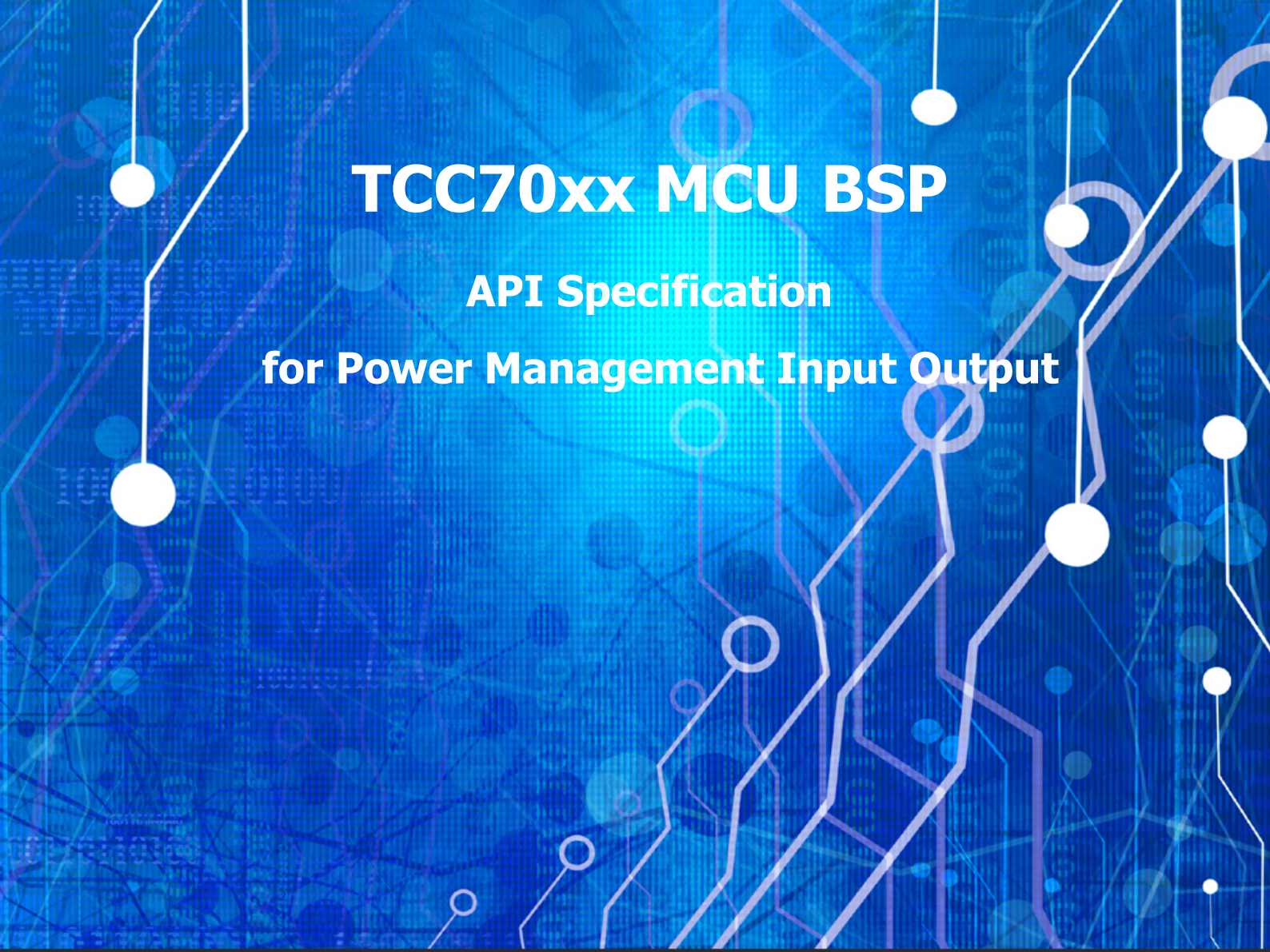




TCC70xx MCU BSP

API Specification

for Power Management Input Output

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the power management input output device driver (hereinafter referred to as PMIO driver) of TCC70xx MCU BSP. For more detailed information on power management input output, refer to Chapter 24 Power Management I/O (PMIO) in "*TCC70xx Full Specification*". [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

PMIO	Power Management Input Output
BU	Battery Unit or B+
RTC	Real Time Clock
ACC	Accessory

3 SOURCE FILES

3.1 Location of Source Files

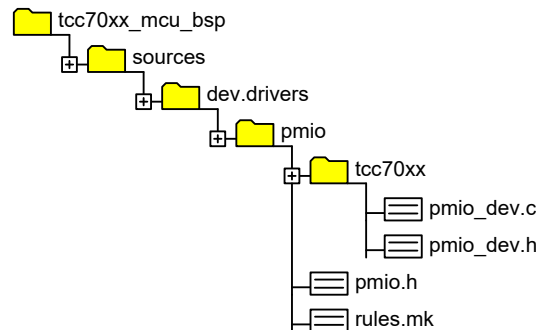


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- pmio_dev.c
 - Power Manager driver source
- pmio_dev.h
 - Power Manager driver header
- pmio.h
 - Power Manager header for functional configuration, parameter define, and interface
- rules.mk
 - Rules used for build

4 ENUMERATIONS

4.1 PMIOMode_t

Enumeration variables to manage power mode. This value is used to perform power-down.

Declaration

```
typedef enum
{
    PMIO_MODE_NOT_SET          = 0,
    PMIO_MODE_POWER_ON         = 1,
    PMIO_MODE_POWER_DOWN       = 2,
    PMIO_MODE_POWER_DEEP_DOWN  = 3,
    PMIO_MODE_MAX               = 4
} PMIOMode_t;
```

Description

Value	Description
PMIO_MODE_NOT_SET	Power mode is not set
PMIO_MODE_POWER_ON	Backup RAM retention off Backup RAM initialization on Backup RAM power on SW_LDO on
PMIO_MODE_POWER_DOWN	Backup RAM retention on Backup RAM initialization off Backup RAM power on SW_LDO off
PMIO_MODE_POWER_DEEP_DOWN	Backup RAM retention off Backup RAM initialization on Backup RAM power off SW_LDO off
PMIO_MODE_MAX	Maximum value of Power mode

Remark

None

4.2 PMIOReason_t

Enumeration variables to manage power-on reason state. This value is used to notify you.

Declaration

```
typedef enum
{
    PMIO_REASON_NOT_SET    = 0,
    PMIO_REASON_BU_ON     = 1,
    PMIO_REASON_ACC_ON    = 2,
    PMIO_REASON_CAN_ON    = 3,
    PMIO_REASON_RTC_ON    = 4,
    PMIO_REASON_MAX       = 5
} PMIOReason_t;
```

Description

Value	Description
PMIO_REASON_NOT_SET	Power-on reason is <u>not</u> set.
PMIO_REASON_BU_ON	Power-on reason is BU.
PMIO_REASON_ACC_ON	Power-on reason is ACC ON.
PMIO_REASON_CAN_ON	Power-on reason is CAN ON.
PMIO_REASON_RTC_ON	Power-on reason is RTC wake up.
PMIO_REASON_MAX	Maximum value of Power-on reason

Remark

None

4.3 PMIOPortSel_t

Enumeration variables to manage GPIO_K port. This value is only used to control the GPIO_K/PCTL port.

Declaration

```
typedef enum
{
    PMIO_PORT_SEL_RELEASE      = 0,
    PMIO_PORT_SEL_GPIO        = 1,
    PMIO_PORT_SEL_PMGPIO_IN   = 2,
    PMIO_PORT_SEL_PMGPIO_OUT_L = 3,
    PMIO_PORT_SEL_PMGPIO_OUT_H = 4,
    PMIO_PORT_SEL_PMGPIO_PU    = 5,
    PMIO_PORT_SEL_PMGPIO_PD    = 6,
    PMIO_PORT_SEL_PCTLIO_IN    = 7,
    PMIO_PORT_SEL_PCTLIO_OUT_L = 8,
    PMIO_PORT_SEL_PCTLIO_OUT_H = 9,
    PMIO_PORT_SEL_PCTLIO_PU    = 10,
    PMIO_PORT_SEL_PCTLIO_PD    = 11
} PMIOPortSel_t;
```

Description

Value	Description
<i>PMIO_PORT_SEL_RELEASE</i>	GPIO_K port with this value selected is controlled by PMIO and is set to the input disable state.
<i>PMIO_PORT_SEL_GPIO</i>	GPIO_K port with this value selected is controlled by GPIO. Use the GPIO driver to control this port.
<i>PMIO_PORT_SEL_PMGPIO_IN</i>	GPIO_K port with this value selected is controlled by PMIO and is set to the input enable state.
<i>PMIO_PORT_SEL_PMGPIO_OUT_x</i>	GPIO_K port with this value selected is controlled by PMIO and is set to the output low/high state. ■ <i>PMIO_PORT_SEL_PMGPIO_OUT_L</i>: Low ■ <i>PMIO_PORT_SEL_PMGPIO_OUT_H</i>: high
<i>PMIO_PORT_SEL_PMGPIO_Px</i>	GPIO_K port with this value selected is controlled by PMIO. ■ <i>PMIO_PORT_SEL_PMGPIO_PU</i>: Pull-up ■ <i>PMIO_PORT_SEL_PMGPIO_PD</i>: Pull-down
<i>PMIO_PORT_SEL_PCTLIO_IN</i>	PCTL port with this value selected is controlled by PMIO and is set to the input enable state.
<i>PMIO_PORT_SEL_PCTLIO_OUT_x</i>	PCTL port with this value selected is controlled by PMIO and is set to the output low/high state. ■ <i>PMIO_PORT_SEL_PCTLIO_OUT_L</i>: Low ■ <i>PMIO_PORT_SEL_PCTLIO_OUT_H</i>: high
<i>PMIO_PORT_SEL_PCTLIO_Px</i>	PCTL port with this value selected is controlled by PMIO. ■ <i>PMIO_PORT_SEL_PCTLIO_PU</i>: Pull-up ■ <i>PMIO_PORT_SEL_PCTLIO_PD</i>: Pull-down

Remark

None

5 CONSTANTS

5.1 PMIO Interrupt Source Number

Definition variables of GIC's External interrupt number for PMIO GPIOK signal detection

Declaration

```
#define PMIO_VA_INTERRUPT_SRC_BU      ((uint32)(GIC_EXT0))
#define PMIO_VA_INTERRUPT_SRC_ACC    ((uint32)(GIC_EXT1))
#define PMIO_VA_INTERRUPT_SRC_CAN    ((uint32)(GIC_EXT2))
#define PMIO_VA_INTERRUPT_SRC_ALL    ((uint32)(0UL))
```

Description

Value	Description
<i>PMIO_VA_INTERRUPT_SRC_[name]</i>	<i>PMIO Interrupt source number.</i> ■ <i>BU is detected by using GIC_EXT0</i> ■ <i>ACC is detected by using GIC_EXT1</i> ■ <i>CAN is detected by using GIC_EXT2</i>

Remark

Refer to Chapter 24 Power Management I/O(PMIO) in “*TCC70xx Full Specification*” about control registers offset in each base address. [1]

6 APIs

6.1 PMIO_EarlyWakeUp

This function is called in *startup.S*, for PMIO default settings in the initial stage of booting before BSP initialization. If early setting is required in user environment, it can be implemented in this way.

Declaration

```
void PMIO_EarlyWakeUp  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.2 PMIO_Init

This function is called from *bsp.c*, and PMIO is initialized in BSP initialization.

Declaration

```
void PMIO_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.3 PMIO_PowerDown

This function is called by interrupt in *pmio_dev.c* and performs power-down when an interrupt to ACC and BU occurs.

Declaration

```
void PMIO_PowerDown
(
    PMIOMode_t tMode
);
```

Parameters

Value	Description
tMode	[in] Power-down mode type. Refer to Chapter 4.1 PMIOMode_t. This function is valid only when the parameter is PMIO_MODE_POWER_DOWN, PMIO_MODE_POWER_DEEP_DOWN among PMIOMode_t.

Return

None

Remark

None

6.4 PMIO_GetBackupRam

Reads the backup RAM register.

Declaration

```
uint32 PMIO_GetBackupRam
(
    uint32 uiAddr
);
```

Parameters

Value	Description
uiAddr	[in] Backup RAM register address

Return

Value	Description
uint32	Backup RAM register value

Remark

None

6.5 PMIO_SetBackupRam

Writes the backup RAM register.

Declaration

```
sint8 PMIO_SetBackupRam  
(  
    uint32 uiAddr,  
    uint32 uiVa  
);
```

Parameters

Value	Description
<i>uiAddr</i>	<i>[in] Backup RAM register address</i>
<i>uiVa</i>	<i>[in] Backup RAM register value</i>

Return

Value	Description
<i>0</i>	<i>Success</i>
<i>1</i>	<i>Failure</i>

Remark

None

6.6 PMIO_GetBootReason

Allows you to distinguish power-on by BU, ACC, CAN, and RTC.

Declaration

```
PMIOReason_t PMIO_GetBootReason
(
    void
);
```

Parameters

None

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

6.7 PMIO_DebugPrintPowerStatus

This API is only for debug.

Declaration

```
void PMIO_DebugPrintPowerStatus  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.8 PMIO_SetGPK

Allows you to use GPIO-K as normal GPIO

Declaration

```
void PMIO_SetGPK
(
    uint32          uiPort32
);
```

Parameters

Value	Description
uiPort32	[in] 32-bit for GPIO_K[18:0] pin Example: GPIO_K_1: uiPort32 = 0x1 GPIO_K_3: uiPort32 = 0x8 or (0x1<<3) GPIO_K_1 and GPIO_K_3: uiPort32 = 0x9

Return

None

Remark

Refer to Chapter 7.7 Set Normal GPK.

7 HOW TO USE PMIO DRIVER

The PMIO driver is initially set by EarlyWakeUp (Pre init) and Init.

7.1 Set Configuration

PMIO driver can be used in configuration flags for power functions. You can see the following flags in *sources/dev.drivers/pmio/pmio.h*.

```
#define PMIO_CONF_DEBUG_ALONE
#define PMIO_CONF_PREVENT_LEAK
#define PMIO_CONF_DET_BU
#define PMIO_CONF_DET_ACC
//#define PMIO_CONF_DET_CAN
//#define PMIO_CONF_RTC_WAKE_UP
//#define PMIO_CONF_MISC_DEV_FUNC
```

A description of each configuration flag is as follows:

Value	Description
PMIO_CONF_DEBUG_ALONE	[on] Print errors and debug log all the time. (<u>Not</u> use a LOG_D/E console.)
PMIO_CONF_PREVENT_LEAK	[on] Prevent power current Leakage when system enters power-down. This setting is applied to Evaluation Board (EVB) provided by Telechips.
PMIO_CONF_DET_BU	[on] BU detection possible through GPIOK port
PMIO_CONF_DET_ACC	[on] ACC detection possible through GPIOK port
PMIO_CONF_DET_CAN	[on] CAN detection possible through GPIOK port
PMIO_CONF_RTC_WAKE_UP	[on] Run RTC Wake-up when retention mode lasts for a certain time. After RTC Wake-up, system enters PMIO_MODE_POWER_DEEP_DOWN immediately. This function requires an RTC driver.
PMIO_CONF_MISC_DEV_FUNC	[on] Run miscellaneous sample test function

7.2 Pre-Initialize PMIO Driver

EarlyWakeUp can be used in ***startup.S*** file as follows when you need faster configuration than BSP initialization.

```
bl  vfp_init
bl  PMIO_EarlyWakeUp
b   cmain
b   .
```

7.3 Initialize PMIO Driver

```
{
    PMIO_Init();
}
```

7.4 Power Down System

```
{
    PMIO_PowerDown(PMIO_MODE_POWER_DOWN);
}
```

7.5 Use BackupRam

```
{
    PMIO_SetBackupRam(PMIO_VA_BACKUP_RAM_ADDR, 1UL);
    PMIO_GetBackupRam(PMIO_VA_BACKUP_RAM_ADDR);
    PMIO_ClearBackupRAM();
}
```

7.6 Get Boot Reason

```
{
    PMIOReason_t reason;
    reason = PMIO_GetBootReason();
    printf("boot reason is %d", reason);
}
```

7.7 Set Normal GPK

```
{  
    PMIO_SetGPK( PMIO_GPK(12)|PMIO_GPK(11) );  
  
    GPIO_Config( GPIO_GPK(12),      ( uint32 ) ( GPIO_FUNC( 1U ) | GPIO_OUTPUT ) );  
    GPIO_Set( GPIO_GPK(12) );  
}
```

8 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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